A New Image Compensation and Segmentation Algorithm to Retrieve Atmospheric Boundary Layer Height for LiDAR RHI 2-D Scan

Le Chen, Zhibin Yu, Shihai Wang, Chunhui He, Lin Mei, Jianchuan Zheng, Zhangjun Wang, Taofeng Gu, and Guoliang Shentu

***Abstract*—The atmospheric boundary layer controls surface–atmosphere energy and matter exchange, and plays a central role in air pollution and weather dynamics. However, conventional vertically pointing LiDAR observations suffer from a near-surface blind zone and cannot provide 2-D information on aerosol distributions, limiting their capability to resolve delicate spatial structures and capture the evolution of the boundary layer. To overcome this limitation, this study employed LiDAR range–height indicator (RHI) scanning to obtain 2-D aerosol observation data with no blind zones and high spatiotemporal resolution. Based on the characteristics of RHI image data, we propose an image compensation and segmentation algorithm (ICSA) for detecting the atmospheric boundary layer height (ABLH). The algorithm compensates for the RHI backscatter signal to more accurately represent aerosol concentration and incorporates Otsu's multithreshold segmentation and connected-component analysis. It offers a new approach to accurately characterize the boundary layer structure and evolution. The proposed algorithm can stably retrieve the ABLH without a predefined height range, and when a predefined height constraint is applied to the gradient method (GM) and STRATfinder methods, the ABLH results from the three methods are generally consistent. Two additional cases further verify the effectiveness of the proposed algorithm in identifying the ABLH during complex boundary layer evolution. The results demonstrate that the method can accurately capture structural variations of the atmospheric boundary layer associated with turbulent mixing and convective transport within the wind field. It also clearly depicts the redistribution process of aerosols under the intermittent turbulent effects induced by near-surface temperature inversions. These cases not only demonstrate the algorithm's exceptional capability to resolve the fine structure of the atmospheric boundary layer but also provide a reliable pathway to deeper insights into the coupled dynamic-thermal mechanisms within the near-surface layer.**

***Index Terms*—Aerosol, atmospheric boundary layer, image segmentation, LiDAR, remote sensing.**

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Le Chen, Zhibin Yu, Shihai Wang, and Chunhui He are with the State Key Laboratory of Solar Activity and Space Weather and School of Aerospace Science, Harbin Institute of Technology (Shenzhen), Shenzhen 518055, China (e-mail: lechen@stu.hit.edu.cn; yuzb@hit.edu.cn; wangshihai@hit.edu.cn; hech@stu.hit.edu.cn).

Lin Mei and Jianchuan Zheng are with Shenzhen National Climate Observatory (Shenzhen Astronomical Observatory), Meteorological Bureau of Shenzhen Municipality, Shenzhen 518040, Republic of China (e-mail: meilin4587@163.com; ynujchzheng@126.com).

Zhangjun Wang is with the State Key Laboratory of Physical Oceanography, Institute of Oceanographic Instrumentation, Qilu University of Technology (Shandong Academy of Sciences), Qingdao, Shandong 266001, China (e-mail: zhangjunwang@qlu.edu.cn).

Taofeng Gu is with Guangzhou Meteorological Comprehensive Guarantee Center, Guangzhou, Guangdong 511430, China (e-mail: gutaofeng@gz.gov.cn).

Guoliang Shentu is with Wuhan Institute of Quantum Technology, Wuhan 430000, China, and also with Shandong Guoyao Quantum Lidar Technology Company Ltd., Jinan 250101, China (e-mail: stgl413@ustc.edu.cn).

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I. Introduction

THE atmospheric boundary layer serves as the crucial interface between the Earth's surface and the free atmosphere, directly influenced by surface dynamics and thermal forcing [1]. The atmospheric boundary layer height (ABLH) is a key parameter characterizing the atmospheric boundary layer, representing the vertical extent of turbulent mixing and the depth of mass, momentum, and energy exchange between the surface and the atmosphere [2], [3], [4]. ABLH plays a decisive role in the dispersion and accumulation of air pollutants [3], [5], [6], [7]. The spatiotemporal variability of the ABLH also modulates the local meteorological and climate system and the surface energy budget [8], [9], [10], [11]. Consequently, accurately measuring ABLH is crucial not only for understanding atmospheric boundary layer processes but also for improving air quality forecasts, numerical weather prediction, aviation safety, and climate assessment.

Currently, various instruments are available for detecting the ABLH. In addition to the classic radiosonde, remote sensing techniques such as GPS, LiDAR, and microwave radar are commonly used [2], [4], [12], [13], [14], [15], [16], [17], [18]. Among these, LiDAR has attracted increasing attention in recent years due to its high spatiotemporal resolution and continuous monitoring capabilities. By detecting the vertical distribution of aerosols, LiDAR can identify the layering structure within the atmospheric boundary layer and, thus,

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determine the ABLH. Numerous studies have confirmed the reliability and robustness of LiDAR-based ABLH retrievals [19], [20], [21], [22], [23], [24], [25]. LiDAR applications have incredibly advanced boundary layer research, leading to significant progress in understanding its structure and evolution. For instance, Su et al. [26] revealed the key mechanism by which aerosols, through heating effects, suppress the entrainment rate and, thus, hinder boundary layer development based on LiDAR observations. Pal and Haeffelin [27] conducted a systematic analysis of the significant seasonal and interannual variations in the atmospheric boundary layer depth over Paris using long-term LiDAR data. Liu et al. [28] conducted joint LiDAR observations with other instruments to analyze the physical properties and sources of aerosols within the nocturnal residual layer over Wuhan.

Most studies of the atmospheric boundary layer using LiDAR employ a laser beam fixed in a vertically pointing observation mode. Due to the hardware limitations, such as detector saturation and geometry overlap, the fixed vertical observation mode has an inherent near-surface blind zone [29], [30]. This blind zone may result in missing observations of the shallow boundary layer and hinder the detection of fine-scale structures. More critically, vertically pointing measurements yield only 1-D columnar profiles, failing to fully capture the boundary layer's horizontal variability and 2-D structure. Consequently, critical spatially heterogeneous processes, such as local convection, shear-driven turbulence, and the coupling of near-surface thermodynamic and dynamic characteristics, cannot be comprehensively captured. This spatial information gap, combined with the near-surface blind zone, constitutes a key constraint in understanding atmospheric boundary layer evolution.

To overcome these limitations, some studies have adopted a range–height indicator (RHI) scanning strategy to obtain the 2-D detailed atmospheric parameters of the atmospheric boundary layer, including atmospheric optical depth, aerosol extinction coefficient, and ABLH [31], [32], [33], [34], [35]. This scanning strategy captures both horizontal and vertical variations near the surface, even within the LiDAR's near-field blind zone, revealing fine-scale structures and spatial heterogeneity of the atmospheric boundary layer. While numerous ABLH identification algorithms have been proposed for 1-D vertical profiles produced by the conventional vertically pointing observations, such as gradient-based methods [22], [36], wavelet covariance methods [37], [38], [39], variance methods [40], [41], and, more recently, image-recognition approaches based on long-term time series [38], [42], [43], [44], [45], [46], [47]. However, none of these methods fully exploits the 2-D spatial information provided by RHI scans, limiting their ability to capture horizontal variability and fine-scale structures.

Aiming to develop a new ABLH identification method for 2-D RHI image data, this study proposes an image compensation and segmentation algorithm (ICSA) that utilizes Otsu's multithreshold segmentation [48], [49] and connected-component analysis techniques [50]. The ABLH obtained by the proposed algorithm is compared with that derived from the classical gradient method (GM) and the STRATfinder algorithm, the latter of which is based on 2-D image analysis [47]. The proposed algorithm exhibits improved performance in complex scenarios, such as those involving cloud and suspended aerosol layers. Three cases are presented to demonstrate the effectiveness of the proposed method based on RHI data. The first case evaluates the algorithm performance, while the other two demonstrate its application in capturing the fine evolution of the near-surface boundary layer. The structure of this article is given as follows. Section II provides a detailed description of the experimental site, instrumentation, and the characteristics of the RHI data. Section III describes the proposed image processing algorithm. Section IV presents the validation experiment and two cases of near-surface boundary layer evolution. Finally, conclusions are presented in Section V.

II. Description of the Site and Instruments

A. Experimental Sites and Equipment

The LiDAR dataset used in this study was collected from the Shiyan Meteorological Observatory at Shenzhen, China (22.65°N, 113.89°E). The observatory is situated in the hinterland of the Pearl River Delta Economic Circle, surrounded by extensive orchards and adjacent to the Tiegang Reservoir in the south, which spans approximately 64 km^2. The reservoir's protected zone is encircled by rapidly expanding urban clusters, making this location an ideal observatory for investigating the urban boundary layer. A 356-m meteorological tower, situated 450 m south of the LiDAR Fig. 1(c), provides reference measurements of conventional meteorological parameters, including wind speed, temperature, and pressure, at multiple levels (10, 20, 40, 50, 80, 100, 150, 160, 200, 250, 300, 320, and 350 m).

The LiDAR employed in this study is a 1550-nm Mie-scattering system mounted on the roof of the laboratory building, as shown in Fig. 1(a). The LiDAR emits a single pulse with an energy of 75 μJ, a radial resolution of 30 m, a temporal resolution of 1 s, and a maximum detection range of 15 km. Due to detector saturation, the near-range signal, within approximately 0.5 km of the LiDAR, becomes unreliable. The LiDAR operates in the RHI scan mode, and the elevation coverage is set from 0° to 75° with a 0.2° step. A single RHI scan takes about 6.5 min. The azimuth of the RHI scan is about 180°, pointing southward, as shown in Fig. 1(b), and primarily covers the vertical atmospheric profile over the reservoir and the orchard, a region with relatively homogeneous subsurface properties. The altitudes above the sea level of the LiDAR and the 356-m meteorological tower are 68.5 and 56 m, respectively, and the altitude difference is added to the meteorological tower base altitude. We used radiosonde data from King's Park, Hong Kong (22°18′43″N, 114°10′22″E), located approximately 47 km from the Shiyan Meteorological Observatory. The altitude of King's Park is 66 m, which is roughly consistent with the height of the LiDAR observations.

B. Characteristics of LiDAR RHI 2-D Images

LiDAR RHI scanning image data show an instantaneous snapshot of the atmosphere in two dimensions, providing a

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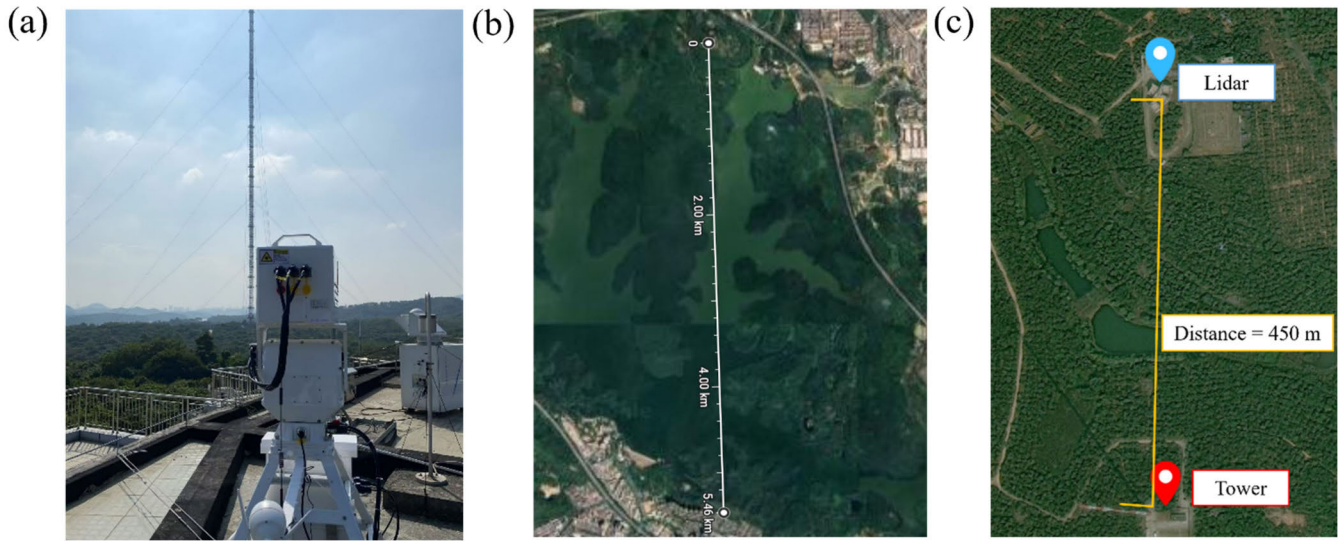


Fig. 1. (a) LiDAR, (b) surface of the scanned area, and (c) distance from the meteorological tower to the LiDAR location (from MAPWORLD, https://map.tianditu.gov.cn/, last access: October 8, 2025).

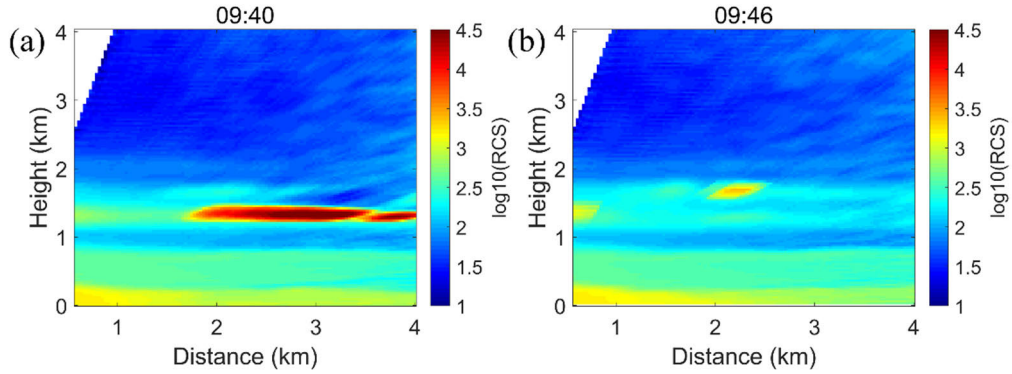


Fig. 2. RHI images at (a) 09:40 and (b) 09:46 on December 26, 2023.

new perspective on the atmospheric boundary layer. Given that the characteristic time scale of atmospheric boundary layer processes is on the order of one hour [1], a single RHI image can be completed in just a few minutes so that the temporal evolution of the atmospheric boundary layer during the scan can be considered negligible. Therefore, the RHI data can faithfully and reliably capture the spatial distribution of the boundary layer in both range and height dimensions.

Fig. 2(a) and (b) shows the RHI images at 09:40 and 09:46 on December 26, 2023, respectively, and the two RHI images clearly illustrate the 2-D horizontal and vertical aerosol distributions. Below 1 km, the aerosol backscattered signal is relatively strong, reflecting higher particle concentrations within the boundary layer. Above 1 km, the signal decreases, corresponding to the free atmosphere with relatively low aerosol concentrations. In Fig. 2(a), a cloud layer is detected at an altitude of approximately 1.5 km with a thickness of about 300 m, and its structure and location are clearly resolved. In contrast, in Fig. 2(b), this cloud layer has moved out of the observational domain, while the boundary layer structure shows little change. Leveraging this 2-D spatial information, significant differences in aerosol distribution between the boundary layer and the free atmosphere can be distinguished, with the sharp signal gradient near 1 km providing a direct basis for determining the boundary layer height. Furthermore, the position and structural characteristics of cloud layers can be accurately identified, reducing the likelihood of misinterpretation. Compared with traditional 1-D observations, the continuous spatial coverage of RHI images substantially enhances the ability to resolve complex boundary layer structures and improves robustness against interference.

Within the atmospheric boundary layer, the RHI images reveal distinct stratifications. Specifically, a near-surface aerosol layer is present below 300 m, where concentrations are markedly higher. These stratifications are captured through gap-free detection enabled by RHI scanning, effectively overcoming the observational blind zones of conventional point-based vertical measurements near the surface. These illustrations demonstrate that LiDAR RHI images enable complete stratification from the ground to the top of the boundary layer to be clearly resolved and potentially provide a fine-grained view of the evolution of the coupling between the boundary layer and the ground surface.

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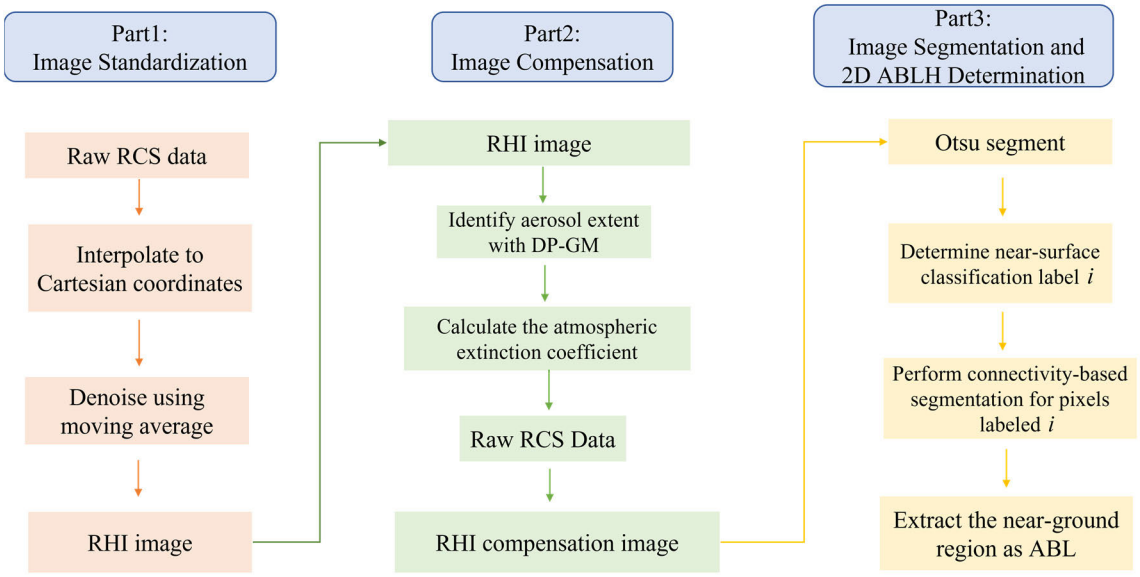


Fig. 3. Flowchart of ICSA for ABLH identification.

III. ICSA for ABLH Determination

A. Algorithm Overview

According to the LiDAR equation, the intensity of the backscattered signal depends on the emitted laser power, the atmospheric extinction coefficient, the range, and other factors. Therefore, preprocessing of the range-corrected signal (RCS) is crucial for generating a standardization RHI scan image. First, the RCS image is interpolated into Cartesian coordinates to standardize the spatial resolution. Then, a 300-m horizontal moving average is applied to reduce random noise. Third, the RCS signal is a function of transmittance, which depends on the atmospheric extinction coefficient and range. Generally, the RCS intensity decreases with increasing range. To ensure that the RHI scan image accurately represents aerosol concentration, atmospheric transmittance compensation is applied to the RCS. After the RHI image is preprocessed, the Otsu method is employed for image segmentation to identify regions with different aerosol concentrations. In addition, the connectivity domain analysis is used to extract the near-surface boundary layer, thereby providing boundary layer height information. The flowchart of the ICSA for ABLH identification is shown in Fig. 3, and specific procedures are presented in Sections III-B–III-D.

B. Image Standardization

To ensure consistency in spatial resolution across multitemporal images, the RCS signals obtained from RHI scans are subjected to a standardization procedure (Part 1 of Fig. 3). The raw 2-D RCS data from RHI scans are stored in polar coordinates, with fixed resolutions in the radial and angular directions. This results in a significant decline in spatial resolution with increasing range, affecting the accuracy of observations of long-distance targets. To eliminate grid distortion caused by uneven sampling and to achieve spatially consistent resolution, the raw polar-coordinate data are interpolated onto a Cartesian coordinate grid. Specifically, a natural interpolation [51] is applied to the inner region of the data, while the boundary regions are handled using nearest-neighbor interpolation [52]. To reduce background noise and improve data quality, a 300-m horizontal sliding-average window is applied, significantly minimizing the impact of random noise.

C. Image Compensation

In the RHI scanning data, due to path-dependent atmospheric attenuation, the RCS exhibits spatially nonuniform decay, as shown in Fig. 2. This leads to differences in RCS intensity within the same height range across different azimuth angles. Therefore, the RCS signal itself is challenging to use for quantitatively characterizing aerosol distribution. In principle, it is necessary to obtain the atmospheric backscatter coefficient, or a proxy proportional to it, to mitigate the impact caused by transmittance differences. Classic backscatter retrieval algorithms, namely, the Klett [54] methods, typically rely on selecting approximately homogeneous reference intervals along a single LiDAR profile as boundary values. However, under RHI observation conditions, the combined effects of cloud layers, residual layers, and spatially inhomogeneous aerosol structures often make it challenging to ensure signal continuity and homogeneity along an individual profile. As a result, determining the reference interval is highly uncertain, affecting the stability and reliability of the retrieval results.

Since a stable reference interval is hard to define for a single profile, directly retrieving a reliable backscatter coefficient under RHI conditions is challenging. To address this, this study develops a transmittance compensation strategy based on a layered mean extinction approximation. It yields an approximated backscatter coefficient $\beta^*(r)$.

This strategy simplifies the radial atmospheric attenuation into two segments. One is an aerosol accumulation region with

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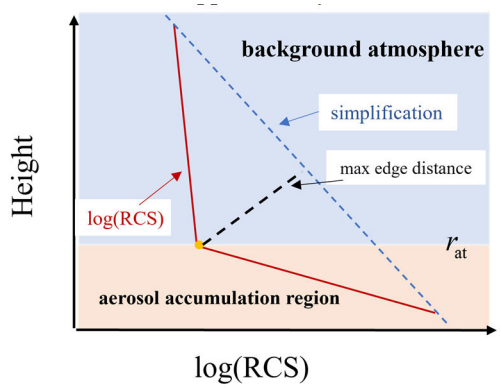


Fig. 4. Schematic of the DP algorithm.

relatively high aerosol loading, and the other is a background atmosphere with relatively low aerosol loading, where molecular scattering and absorption dominate. The layered mean extinction approximation assumes that a single extinction level represents each segment, approximating its average attenuation strength. The LiDAR RCS can be expressed as

$$RCS\,(r) = C\beta\,(r)\exp\left\{-2\int_{r_0}^{r}\left[\sigma_a\left(r'\right)+\sigma_m\left(r'\right)\right]dr'\right\} \quad (1)$$

where r denotes the radial distance from the LiDAR, expressed as $r = (x^2 + y^2)^{1/2}$. β represents the atmospheric backscatter coefficient, while σ_a and σ_m denote the aerosol and molecular extinction coefficients, respectively. As we discussed above, the accurate solution for σ_a and σ_m is impossible for a single-wavelength elastic LiDAR RHI scan measurement. To reduce the influence of path-dependent attenuation, the transmittance compensated backscatter coefficient $\beta^*(r)$ is expressed as follows:

$$\beta^*\,(r) = RCS\,(r)\cdot\exp\left\{2\int_{r_0}^{r}\left[\hat{\sigma}_t\left(r'\right)\right]dr'\right\} \propto \beta\,(r) \quad (2)$$

where $\hat{\sigma}_t$ is the equivalent total extinction coefficient used for compensation. Under the two-layer assumption, $\hat{\sigma}_t$ is expressed in a piecewise form

$$\hat{\sigma}_t = \begin{cases}\bar{\sigma}_a + \sigma_m, & r \le r_{at}\\ \sigma_m, & r > r_{at}\end{cases} \quad (3)$$

where $\bar{\sigma}_a$ is the representative aerosol extinction within the aerosol accumulation region and r_{at} its top. r_{at} is estimated to delineate the aerosol accumulation region. Within this region, the Fernald method is applied to obtain the extinction coefficient profile [53], [54], which is then averaged over the aerosol accumulation region to derive $\bar{\sigma}_a$. In the background atmosphere above r_{at}, aerosol extinction is assumed negligible, and the extinction is approximated by the molecular term σ_m, which is estimated using an empirical atmospheric model [55].

To estimate r_{at}, we introduce a two-step procedure that combines the Douglas–Peucker (DP) algorithm and the GM, forming the DP–GM algorithm. First, we perform horizontal averaging on signals with similar horizontal distribution patterns in the 2-D image to obtain a 1-D cumulative profile. This processing approach effectively enhances the signal-to-noise ratio and suppresses noise. Then, the DP algorithm is applied to the cumulative profile to obtain a coarse estimate of r_{at} [56], [57]. In an ideal uniform atmosphere, the logarithmic slope of RCS should be constant, as shown by the red solid line in Fig. 4 [58]. The lower and upper segments of the red solid line represent the aerosol accumulation region and background atmosphere, respectively. The inflection point between the aerosol accumulation region and background atmosphere, which is the yellow point in Fig. 4, is located farthest from the line connecting the first signal in the aerosol accumulation region and the last signal in the background atmosphere, as shown with the blue dashed line in Fig. 4. The DP algorithm effectively smooths the cumulative vertical profile composed of linear segments by reducing the number of points. The simplified vertical profile preserves the rough shape of the original curve but consists only of a subset of the points that defined the original curve. This coarse localization step provides a reasonable search range for the subsequent GM, avoiding blind searches while suppressing interference from clouds and noise. Finally, the constrained 1-D cumulative profile is subjected to fine localization using traditional GM to estimate r_{at}.

To demonstrate the practical effectiveness of the DP–GM algorithm for identifying r_{at}, three typical cumulative vertical profiles are used to test the two-step procedure. For the general case shown in Fig. 5(a), after applying the inflection point extracted by the DP algorithm as a height constraint, the GM successfully identifies r_{at}. For the vertical profile with a second layer around 1 km, as shown in Fig. 5(b), although the position extracted by the DP algorithm was slightly higher than the actual inflection point, the GM still correctly locates r_{at} within the constrained range. For the vertical profile interfered with by a cloud layer around 3 km, as shown in Fig. 5(c), the DP–GM successfully identifies r_{at}, which is shown by the green star mark, while the GM without the DP algorithm misclassifies cloud layers as r_{at}, shown by the black star mark. Therefore, introducing height limits provided by the DP algorithm enables the GM to identify r_{at}, suppressing cloud and noise interference. Additional results for two continuous observation periods, highlighting the robustness of DP–GM under cloud interference, are presented in Section II of the Supplementary Material.

It should be noted that r_{at} is derived from the horizontal accumulation of RCS, which is a form of spatial statistical aggregation. It characterizes the overall consistency of high aerosol concentration features across different horizontal locations, rather than the true vertical structure at any single location. Accordingly, r_{at} should be interpreted as a statistical transition height at which aerosol spatial coherence weakens significantly. It is not equivalent to the ABLH, which is governed by dynamical and thermodynamical processes.

Using data from 09:46 on December 26, 2023, as an example, we demonstrate the process of transmittance compensation. After applying the transmittance correction using (2), the attenuated signals at farther ranges are effectively compensated. As shown in Fig. 6, the horizontal inhomogeneity of atmospheric structures caused by transmittance decay is corrected, and the stratified structure of the boundary layer becomes more clearly defined. After compensation, the polar

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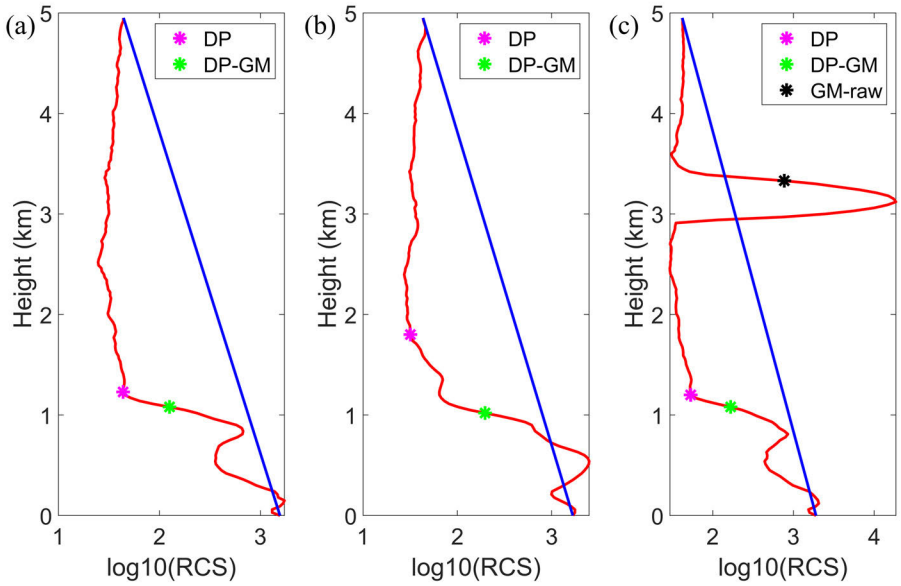


Fig. 5. r_{at} identification using the combined DP–GM method. Purple asterisks indicate r_{at} identified by the DP method, black markers represent r_{at} identified by the GM method without height constraints, and green markers denote r_{at} determined by the combined DP–GM approach. (a)–(c) are the r_{at} identifications for three different cases.

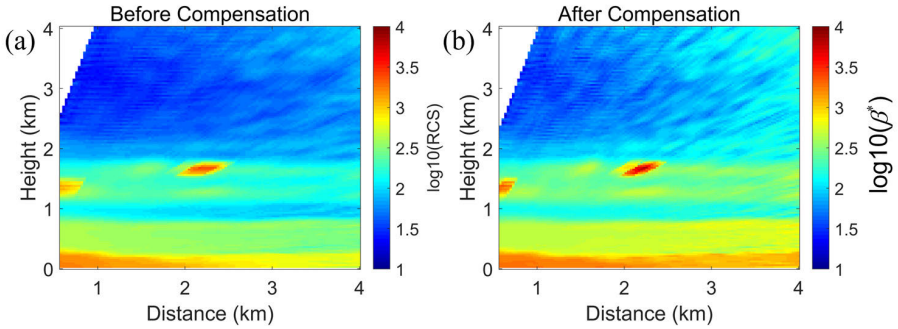


Fig. 6. RHI images at 09:46 on December 26, 2023 (a) before and (b) after transmittance compensation.

sampling pattern becomes more apparent (e.g., near the cloud layer at ~1.8 km), which is a processing-induced artifact. As this effect mainly occurs at longer ranges, it does not affect the identification of the ABLH.

D. Image Segmentation and 2-D ABLH Identification

Aerosol concentrations can be clearly distinguished through variations in RHI image intensity. This spatial contrast in brightness provides a unique advantage for identifying atmospheric structures such as the boundary layer, the residual layer, the cloud layers, and the advected aerosol layers. In RHI images, different atmospheric structures often exhibit distinct peaks in their histogram distributions. Given the characteristics of RHI images, the Otsu multilevel thresholding method can be applied to achieve adaptive segmentation.

The core principle of the Otsu method is to identify an optimal set of thresholds that either maximizes the between-class variance or, equivalently, minimizes the within-class variance. The magnitude of variance reflects the uniformity of the intensity distribution. Minimizing within-class variance ensures that pixel intensities within the same class are highly concentrated, thereby guaranteeing that each segmented region exhibits internal uniformity. Concurrently, maximizing the between-class variance ensures that the intensity differences between distinct classes (i.e., different structures) are sufficiently pronounced, thus achieving clear separation between disparate structures.

Assume the image intensities range $I(M, N)$ from 0 to L −1, and the total quantity of pixels is $M * N$. The probability distribution of a histogram p_i can be represented as

$$p_i = \frac{h(i)}{M * N} \tag{4}$$

where $h(i)$ is the quantity of pixels with intensity value i.

Assuming that $K - 1$ thresholds $(t_1, t_2, \ldots, t_{K-1})$ divide the image into K classes: C_1 for $\{0, \ldots, t_1\}$, C_2 for $\{t_1 + 1, \ldots, t_2\}, \ldots,$ and C_K for $\{t_{K-1} + 1, \ldots, L - 1\}$. The mean value μ_j and the probability ω_j of the class C_j pixels are represented as

$$\mu_j = \frac{1}{\omega_j} \sum_{i \in C_j} i \cdot p_i \tag{5}$$

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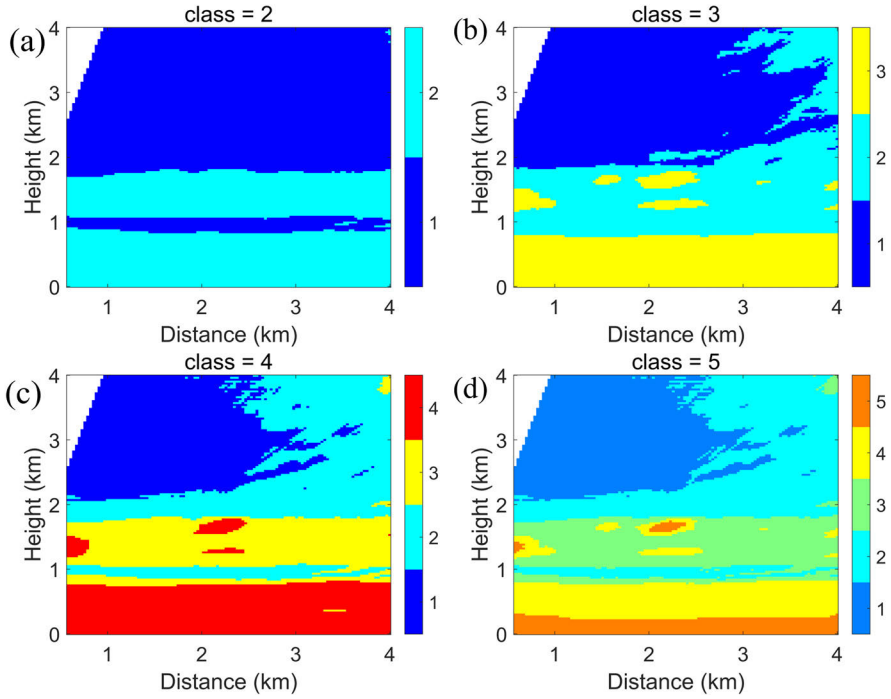


Fig. 7. Classification results of the RHI image at 09:46 on December 26, 2023, using the multilevel Otsu method. (a) Classification into two classes, (b) into three classes, (c) into four classes, and (d) into five classes.

$$\omega_j = \sum_{i \in C_j} p_i. \tag{6}$$

The total interclass variance σ_B^2 can be represented as

$$\sigma_B^2 = \sum_{j=1}^{k-1} \omega_j \left(\mu_j - \mu_T\right)^2 \tag{7}$$

where μ_T is the total mean value of all class pixels and is expressed as

$$\mu_T = \sum_{i=0}^{L-1} i \cdot p_i. \tag{8}$$

The objective of image segmentation is to find the optimal thresholds $(t_1^*, t_2^*, \ldots, t_{K-1}^*)$ such that

$$\left(t_1^*, t_2^*, \ldots, t_{K-1}^*\right) = \arg \max_{t_1, \ldots, t_{k-1}} \sigma_B^2. \tag{9}$$

The search-based optimization algorithm with the nelder-mead method is used to find the optimal threshold.

Fig. 7 illustrates the classification results under different class numbers. When the number of classes is too low, distinct atmospheric structures tend to be merged, resulting in poor separability. Conversely, an excessive number of classes allows for finer representation of the vertical atmospheric structure but significantly increases computational complexity. The atmosphere contains background atmosphere, residual layers, boundary layers, and other structures with distinct intensity differences. Therefore, at least three classes are recommended. When the atmosphere is more stratified, such as with clouds, the number of classes should be increased further to ensure clear stratification.

The ABLH, located near the surface, exhibits uniform intensity, making it easily identifiable in the segmented image. The atmospheric boundary layer type is defined by the mode of first row classification, which is represented as

$$C_g = \text{mode}\left(\{S\left(1, c\right) | c = 1, 2, \ldots, N\}\right) \tag{10}$$

where S is the segmented label matrix of the image and $S(r, c) \in \{C_1, C_2, \ldots, C_K\}$.

For all pixels satisfying $\{(r, c) | S(r, c) = C_g\}$, the image is divided into V connected regions by analyzing the connectivity of their octant neighborhoods, denoted as $\mathfrak{R} = \{R_1, R_2, \ldots, R_V\}$. To avoid interference from noise, regions containing fewer than 20 pixels are defined as noise and are excluded from $\mathfrak{R}$. The set of regions $\mathfrak{R}$ may comprise various atmospheric structures, such as the atmospheric boundary layer, residual layer, or clouds, as presented by the red areas in Fig. 7(c).

The subregion R_a, which includes the first row, is defined as the atmospheric boundary layer. The ABLH is defined as the maximum value of height $H_a(r, c)$ of R_a

$$\text{ABLH}\left(c\right) = \max_{r \in R_{a(c)}} \left\{H_a\left(r, c\right) | \left(r, c\right) \in R_a\right\}. \tag{11}$$

IV. Evaluation of ICSA and Case Studies

This section utilizes a dataset comprising challenging atmospheric boundary layer cases, including the presence of cloud layers and low ABLHs, to assess the accuracy and robustness of the ICSA. Comparisons with the conventional GM and the emerging STRATfinder method demonstrate the superiority of ICSA for boundary layer identification from 2-D spatial imagery. Subsequently, the algorithm is applied to two representative cases to illustrate its capability to capture the evolution of boundary layer structures under dynamic and thermodynamic processes.

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A. Verification of ICSA

A 24-h continuous observational dataset on December 30, 2023, is chosen because the atmospheric boundary layer evolution was relatively complex and challenging. Multilayered clouds above 1.5 km were the primary source of interference in identifying the ABLH from LiDAR data.

To evaluate the performance of the proposed algorithm, the ICSA algorithm is compared with two existing ABLH retrieval methods, namely, the GM and the STRATfinder algorithm. The GM is a classical and well-established ABLH detection technique that has been widely applied in numerous studies and is commonly used as a reference for evaluating the performance of new algorithms [17], [18], [46], [60]. In recent years, 2-D continuity-based image methods for boundary layer identification have been continuously developed, among which the STRATfinder family of algorithms has demonstrated strong performance based on continuous boundary layer imagery [38], [43], [45], [47], [61].

In addition, the ABLH values retrieved from radiosonde observations launched at 08:00 and 20:00 local time at the King's Park station in Hong Kong are introduced as an independent reference to assess the retrieval performance of the proposed ICSA method.

The GM calculates the gradient in the first derivative of the RCS, and the ABLH is defined by the altitude where the gradient reaches the maximum negative, given by (10)

$$H = \min_r \left\{ \frac{dS}{dr} \right\} = \min_r \left\{ \frac{d \ln [\mathrm{RCS}(r)]}{dr} \right\} \tag{12}$$

where H denotes the identified boundary layer height. The GM offers computational simplicity, but taking the first derivative inherently amplifies noise, making the method highly sensitive to it. To mitigate the effect of noise, a moving average filter is applied to the first-order derivative to suppress random noise. Nevertheless, it should be noted that such smoothing is ineffective in the presence of clouds or multiple aerosol layers [62] and should, therefore, be used with caution.

As a representative algorithm based on 2-D continuity imagery, the STRATfinder algorithm proposed by Kotthaus et al. [47] is adopted for comparative analysis in this study. The algorithm formulates the identification of the ABLH as a shortest-path problem in a 2-D image. A Dijkstra-based [63] search is then applied under continuity constraints to determine an optimal path with the minimum total cost, enabling robust identification of ABLH. In the original STRATfinder framework, the path search is conducted in time–height 2-D imagery, with the core assumption that the ABLH evolves continuously in time.

In this study, the continuity assumption is treated as a 2-D spatial-domain problem rather than a temporal-domain problem for the RHI LiDAR image. Specifically, the time–height coordinate system is mapped to a horizontal–vertical coordinate. It reformulates the assumption of temporal continuity of the boundary layer height as spatial continuity, enabling continuous tracking of boundary layer structures along the horizontal direction. In the implementation, the RHI data have a grid spacing of 30 × 30 m in the horizontal and vertical directions. A slope constraint is imposed between adjacent pixels, with a threshold value of 1.5. When the ratio of the height difference between neighboring nodes to the corresponding horizontal distance exceeds this threshold, the connection is considered inconsistent with the spatial continuity assumption and is excluded during the shortest-path search.

In constructing the path cost function, only the vertical gradient of the signal is considered. The variance-based weighting of the STRATfinder algorithm is omitted because the boundary layer top shows very little variance in the lake environment. The cost function is defined as follows [47]:

$$C = \left[\frac{\partial \left[\log 10 (X) \right]}{\partial z} \right]^{-1}. \tag{13}$$

We compared the ABLH derived from the ICSA, GM, and STRATfinder with radiosonde observations at 08:00 and 20:00 local time (see Fig. 8). The radiosonde-derived ABLH (H_{RS}) is obtained using the potential temperature method proposed by Liu and Liang [15]. According to their classification, 08:00 and 20:00 represent neutral and stable atmospheric conditions, respectively. At 08:00 [see Fig. 8(a) and (b)], the radiosonde ABLH was 348 m, while the ABLH values estimated by the ICSA, GM, and STRATfinder were 480, 570, and 630 m, respectively. All three algorithms overestimated the ABLH, with ICSA showing the highest consistency with the radiosonde data. At 20:00 [see Fig. 8(c) and (d)], the radiosonde ABLH was 86 m, while the ICSA, GM, and START yielded 210, 120, and 300 m, respectively. Notably, the GM and STRAT required predefined altitude constraints to mitigate cloud interference, whereas ICSA independently identified the ABLH without prior information.

The ABLH values derived from the ICSA, GM, and STRATfinder methods were all higher than those measured by the radiosonde. This discrepancy may be partly attributed to the differing detection principles: LiDAR identifies the ABLH based on aerosol distribution, while the radiosonde determines it using the potential temperature method. Another contributing factor may be differences in observation environments and the approximately 47-km spatial separation between the LiDAR and radiosonde stations.

Fig. 9 presents the 2-D spatial distribution of the ABLH at different times on December 30, 2023 (08:05, 10:54, 14:01, 17:22, 19:39, and 23:25). The background color denotes the proxy backscatter coefficient intensity, while the red stars, gray dots, and black squares indicate the ABLH identified by the ICSA, GM, and STRATfinder methods, respectively. Because multilayer cloud structures were present above 1.5 km on that day [see Fig. 9(d)], their strong backscatter signals could interfere with the GM and STRATfinder methods' identification of the ABLH. Therefore, the search height range for these two methods was restricted to below 1.5 km. This constraint does not affect the identification of the ABLH, as the maximum observed ABLH on that day was approximately 1 km. As shown in Fig. 9, all three methods capture the overall evolution of the ABLH throughout the day, with consistent diurnal variation. Around 14:01 local time, the ABLH increases markedly [see Fig. 9(c)],

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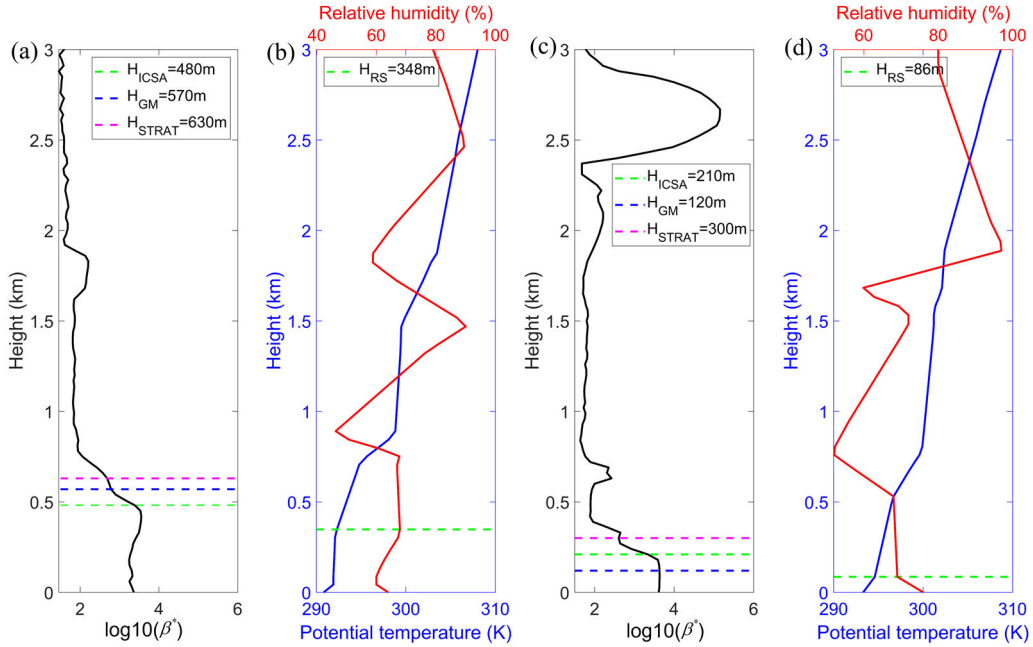


Fig. 8. (a) and (c) LiDAR approximates backscatter coefficient profile (black solid lines). The green dashed line denotes the ABLH derived from the ICSA algorithm, the blue dashed line from the GM method, and the magenta dashed line from the STRATfinder method. (b) and (d) Radiosonde profiles of potential temperature (blue solid lines) and relative humidity (red solid lines), with the green dashed line indicating the ABLH obtained from the potential temperature method. (a)–(b) 08:00 on December 30, 2023. (c)–(d) 20:00 on the same day.

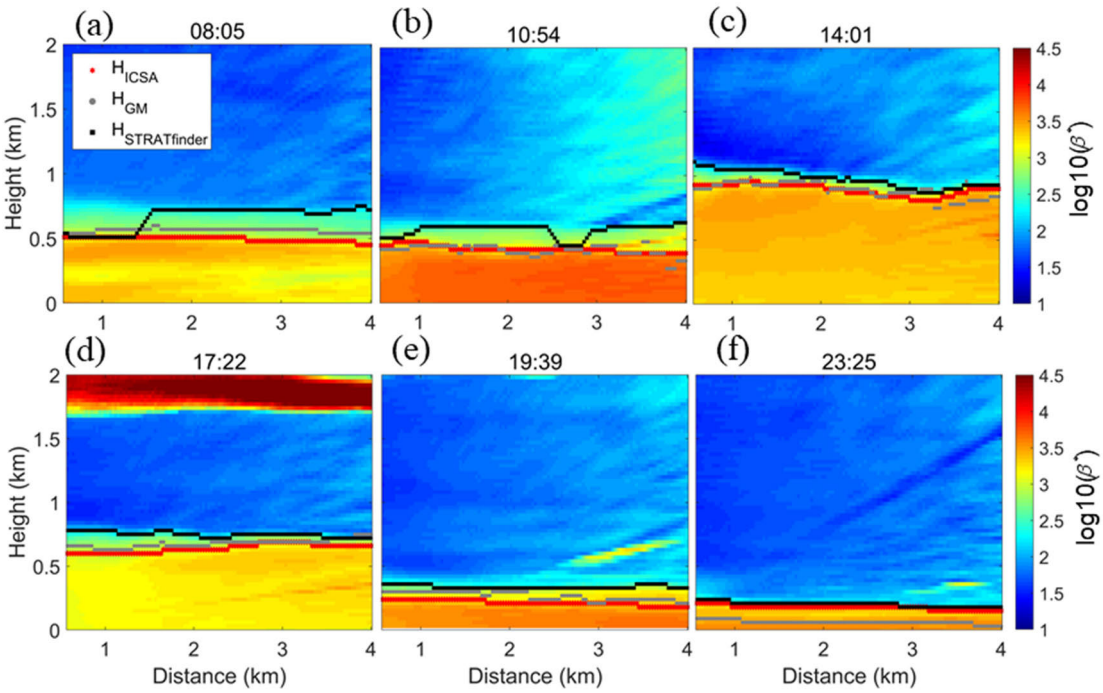


Fig. 9. Snapshots of the spatial distribution of ABLH on December 30, 2023, obtained by the ICSA (red star), GM (gray circle), and STRATfinder (black square). (a)–(f) are the snapshots at different times.

whereas, at 19:39 and 23:25 [see Fig. 9(e) and (f)], the ABLH remains relatively low and shows little variability. Despite overall consistency in the diurnal trend, differences among the methods persist during certain periods. In Fig. 9(a) and (d), the ABLH retrieved by the ICSA and GM methods shows good overall agreement and exhibits stable behavior along the horizontal direction. The STRATfinder method yields slightly higher ABLH and shows modest horizontal variability. Under stable nighttime boundary layer conditions [see Fig. 9(f)], the ABLH estimates derived from the ICSA and STRATfinder methods are generally consistent, while the GM method yields relatively lower ABLH estimates, which may be associated with subtle structures within the stable boundary layer.

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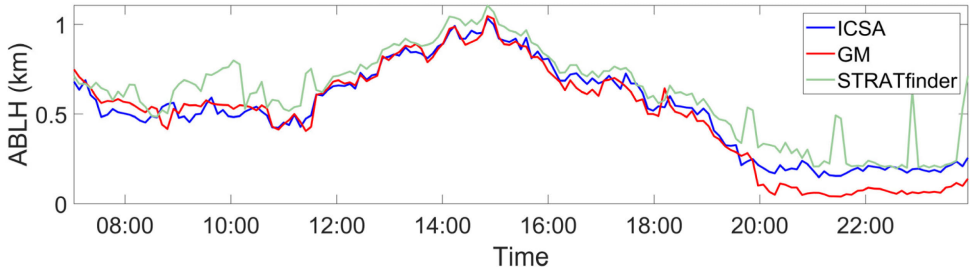


Fig. 10. Temporal evolution of ABLH on December 30, 2023, obtained by the ICSA (blue solid line), GM (red solid line), and STRATfinder (green solid line).

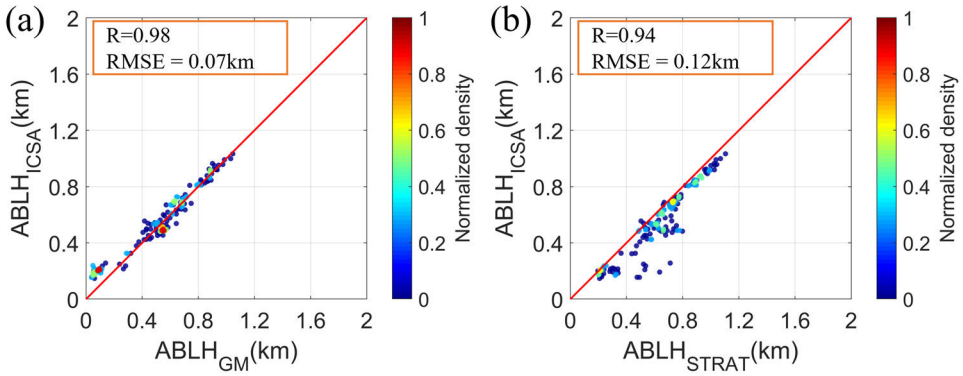


Fig. 11. Scatter plot comparison of ICSA-identified ABLH against (a) GM and (b) STRAT (December 30, 2023). The color of the points represents the scatter normalized density.

Fig. 10 shows the temporal evolution of the ABLH on December 30, 2023, where each time's ABLH is obtained by averaging the corresponding spatial retrievals. Overall, all three methods can represent the diurnal variation in boundary layer height. Before 20:00, the ABLH estimates obtained from all three methods are in close agreement. After 20:00, differences among the methods become apparent. The GM method yields lower ABLH, which is associated with subtle structures within the nighttime atmospheric boundary layer. The ABLH retrieved by the STRATfinder method is generally consistent with that obtained by the ICSA method although occasional scattered values are observed. Fig. 11 presents a statistical comparison of ABLH results on December 30, 2023, obtained by aggregating ABLH across all spatial locations and observation times. The color of the points indicates the density of the samples. Specifically, Fig. 11(a) compares the ABLH estimates derived from the ICSA and GM methods across all aggregated samples, showing strong agreement with a correlation coefficient of 0.98 and a root mean square error (RMSE) of 0.07 km. Fig. 11(b) compares the ICSA and STRATfinder methods using the same sample set, yielding a correlation coefficient of 0.94 and an RMSE of 0.12 km, indicating slightly lower correlation. These statistical results indicate that when manual constraints are introduced to limit the ABLH's search range, the ICSA method shows good overall agreement with both the GM and STRATfinder methods.

Considering the results shown in Figs. 9–11, it can be seen that under a unified constraint on the boundary layer height search range, the ABLH retrieved by the ICSA, GM, and STRATfinder methods exhibits a high level of overall consistency, while noticeable differences still occur during certain periods and at specific spatial locations. These differences are closely related to the spatially inhomogeneous distribution of the atmosphere. In single-time 2-D spatial distribution images, the absence of ABLH temporal evolution constraints allows aloft structures to locally influence the spatial identification of ABLH, leading to relatively higher ABLH estimates at certain locations.

Under this analysis context, the identification of the ABLH relies more strongly on the stability and consistency of spatial criteria. The ICSA is specifically designed for 2-D spatial distribution fields, and its criteria are based on the global statistical characteristics of the 2-D backscatter intensity field and spatial topological connectivity, enabling a stable representation of the 2-D distribution of the boundary layer height under spatially heterogeneous conditions.

B. Atmospheric Boundary Layer Identification Under the Influence of Wind

LiDAR observations in RHI mode can effectively depict the evolution of the 2-D atmospheric structure, offering a powerful tool for investigating atmospheric dynamical processes. Fig. 12 shows the temporal evolution of the 2-D atmosphere from the RHI scan on December 24, 2023. At 15:06 local time, the ABLH was approximately 800 m, with a uniform intensity within the atmospheric boundary layer, indicating that aerosols were well-mixed. Slight fluctuations were observed at the top of the boundary layer [see Fig. 12(a)]. By 17:11 in the evening, the top of the atmospheric boundary layer remained at

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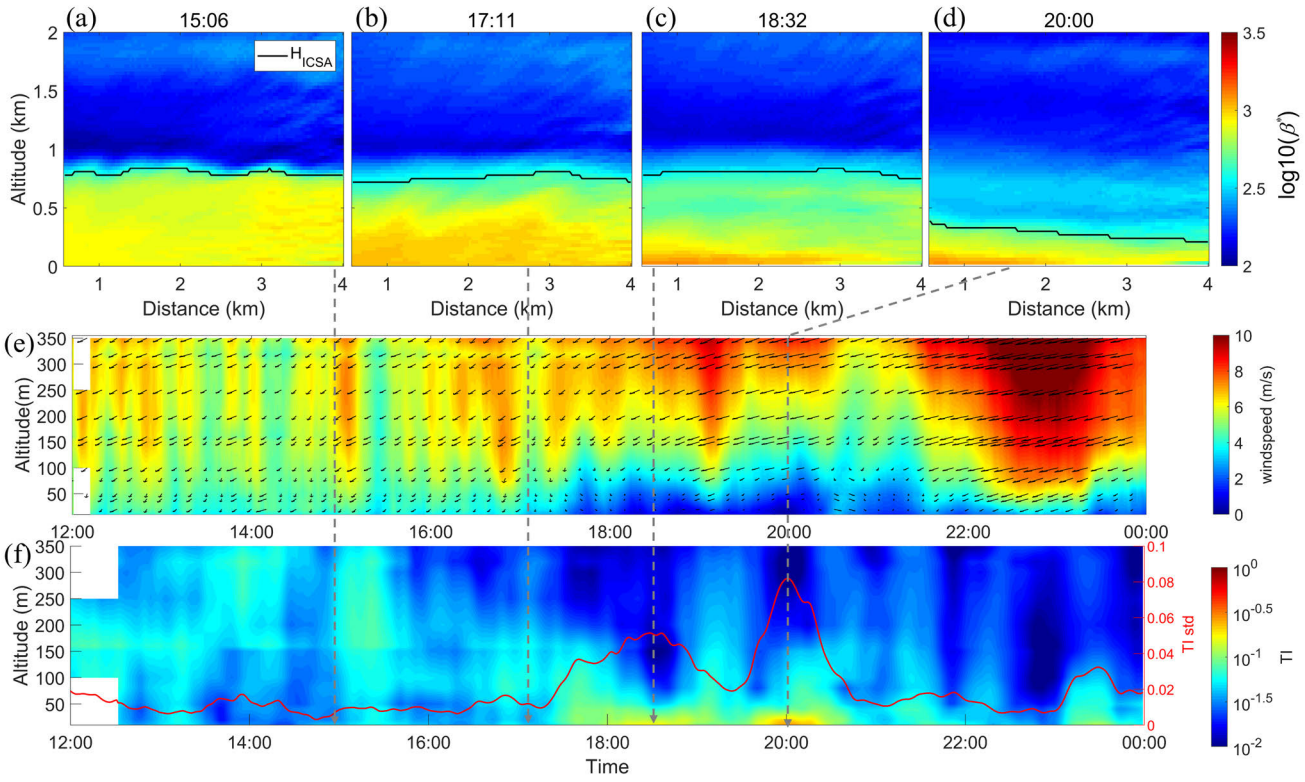


Fig. 12. (a)–(d) Spatial distribution of aerosols from LiDAR observations and the atmospheric variables on December 24, 2023 (e) wind speed (color shading) and direction (arrows) and (f) TI (color shading) and its standard deviation (red solid line). The black solid lines in (a)–(d) represent the ABLH identified by ICSA, and the gray dashed lines indicate the time of the LiDAR images.

approximately 800 m. Pronounced fluctuations were observed between altitudes of 500 and 600 m, particularly at horizontal positions of 1.5 and 3 km [see Fig. 12(b)]. By 18:32, as shown in Fig. 12(c), the fluctuation intensity within the atmospheric boundary layer had slightly weakened, descending to 300–500 m, while the overall aerosol concentration within the boundary layer had also decreased. The atmospheric boundary layer tended to exhibit stratified structures, characterized by a thin residual layer at 600 m and a surface layer (~50-m thick) with elevated aerosol concentrations. Between 50 and 500 m, aerosols were distributed heterogeneously, with concentrations generally decreasing with increasing altitude. By 20:00, as shown in Fig. 12(d), the atmospheric boundary layer top decreased to about 300 m. The aerosol concentration in the residual layer at 600 m had diminished significantly, while the concentration in the surface layer remained relatively high.

Fig. 12(e) displays the wind speed (color shading) and wind direction (arrows) obtained from the 356-m meteorological tower, while the colormap in Fig. 12(f) shows the turbulence intensity (TI), calculated from anemometer wind speed measurements [64]. The red line in Fig. 12(f) is the standard deviation of the vertical distribution of the TI at a given time. During 12:00–17:00, due to surface heating-induced convection, the TI was maintained at 0.05–0.07, with a vertically uniform profile [standard deviation <0.02; see Fig. 12(f)]. Concurrently, the wind field was dominated by northeasterly winds with speeds of 4–6 m/s, exhibiting overall spatial homogeneity [see Fig. 12(e)]. At this time, relatively strong turbulence maintained slight fluctuations at the top of the atmospheric boundary layer. After 17:00, TI at 200–350 m rapidly decreased to below 0.03, reducing aerosol upward diffusion capacity and consequently lowering the boundary layer height. Meanwhile, the wind field underwent significant changes, with the wind direction shifting from northeasterly to northwesterly. The wind speed dropped sharply to 1 m/s near the surface (<100 m), whereas, above 200 m, the wind remained northeasterly with speeds exceeding 7.5 m/s. This establishes strong vertical shear, which induces Kelvin–Helmholtz waves and causes pronounced aerosol fluctuations, as shown in Fig. 12(b).

By 18:30, reduced thermal radiation confined turbulence to below 100 m near the surface [18:00–18:40; see Fig. 12(f)], leading to a descent of the atmospheric boundary layer. Near-surface aerosol concentrations increased and partially remained, forming a residual layer. About 20:00, vertical shear weakened, and the atmospheric boundary layer tended toward a stable state. At this time, wind speeds increased significantly, enhancing horizontal advection and leading to gradual dispersion and weakening of aerosols in the residual layer. Wind speed at 350 m reached 9 m/s, and extrapolation based on geostrophic balance indicates even stronger winds at 600 m, further highlighting the contribution of upper-level strong winds to advection.

These observations reveal that wind plays a key role in the evolution of aerosol stratification in the atmospheric boundary

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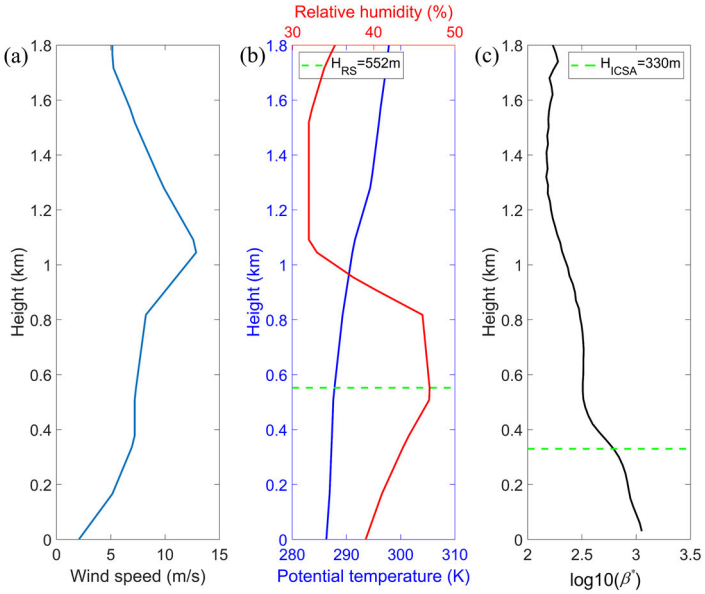


Fig. 13. (a) Radiosonde wind speed profile, (b) potential temperature (blue solid line) and relative humidity (red solid line) profiles, and (c) LiDAR backscatter profile at 20:00 local time on December 24, 2023. The green dashed lines in (b) and (c) indicate the ABLH derived from the radiosonde and from the LiDAR using the ICSA method, respectively.

layer. During the daytime, thermal turbulence facilitates sufficient mixing of aerosols within the atmospheric boundary layer. In the evening, as thermal turbulence weakens, the wind's contribution to the atmospheric boundary layer's evolution becomes more apparent, particularly under strong wind conditions. Specifically, turbulence within the atmospheric boundary layer is generally weak and tends to exhibit laminar characteristics, with aerosols in the upper boundary layer mainly transported by advection. Although the presence of wind shear can trigger Kelvin–Helmholtz waves, it does not evolve into turbulence strong enough to enhance vertical diffusion. In the near-surface layer, strong turbulence is generated by surface friction. However, because it was confined near the surface and could not extend upward, aerosol concentrations at ground level remained substantially higher than those in the upper layers.

We compared the ABLH derived from the ICSA method with that obtained from the radiosonde (see Fig. 13). At 20:00 local time on December 24, 2023, the radiosonde-derived ABLH was 552 m [see Fig. 13(b)], while the ICSA-derived ABLH was 330 m [see Fig. 13(c)], indicating that ICSA underestimated the ABLH. This underestimation is attributed to the strong advection associated with high wind speeds, which transports aerosols outward and reduces their concentration aloft. As a result, the aerosol-based LiDAR retrieval yields a lower ABLH compared with the potential-temperature-based radiosonde method.

C. Atmospheric Boundary Layer Identification Under the Evolution of the Inversion Layer

Fig. 14 illustrates the multilayer evolution of aerosols under stable conditions during the night of January 20, 2025. The black solid lines in (a)–(d) represent the ABLH identified by ICSA. At 22:31, the ABLH was approximately 1.4 km. Aerosols were distributed relatively uniformly within the layer, except for elevated concentrations below 200 m near the surface [see Fig. 14(a)]. From 22:31 to 23:59, the ABLH stabilized around 1 km, while significant multilayer evolution occurred within the atmospheric boundary layer. By 22:56, stratification began around 500 m. Two aerosol layers formed, one between 500 m and 1 km, and another below 400 m, with aerosol concentrations remaining higher below 200 m [see Fig. 14(b)]. By 23:34, the stratification initially below 400 m had gradually extended to 600 m, accompanied by a more uniform aerosol distribution within the layer. At the same time, aerosol concentrations between 600 m and 1 km decreased [see Fig. 14(c)]. By 23:59, three distinct aerosol layers were observed. The near-surface layer dropped below 90 m, a new middle layer appeared between 500 and 600 m, and a weaker upper layer formed between 800 m and 1 km [see Fig. 14(d)].

The atmospheric boundary layer is commonly regarded as stable and homogeneous [65]. However, this observation indicates that even when the boundary layer height remains unchanged, internal stratifications can still develop. The evolution of the near-surface aerosol layer in the nocturnal atmospheric boundary layer was analyzed using temperature data from 13 levels on the meteorological gradient tower. Fig. 14(e) shows the temperature distribution and the variation of visibility. Before 23:00, a strong temperature inversion persisted below 250 m, while, after 23:00, the inversion between 100 and 250 m began to break up and dissipate. The temperature inversion below 100 m remained stable throughout the period. In Fig. 14(e), the black solid line denotes visibility at 50 m, and the purple solid line denotes visibility at 300 m. At 50 m, visibility remained nearly constant at about 10 km. At 300 m, visibility remained near 20 km before 23:10,

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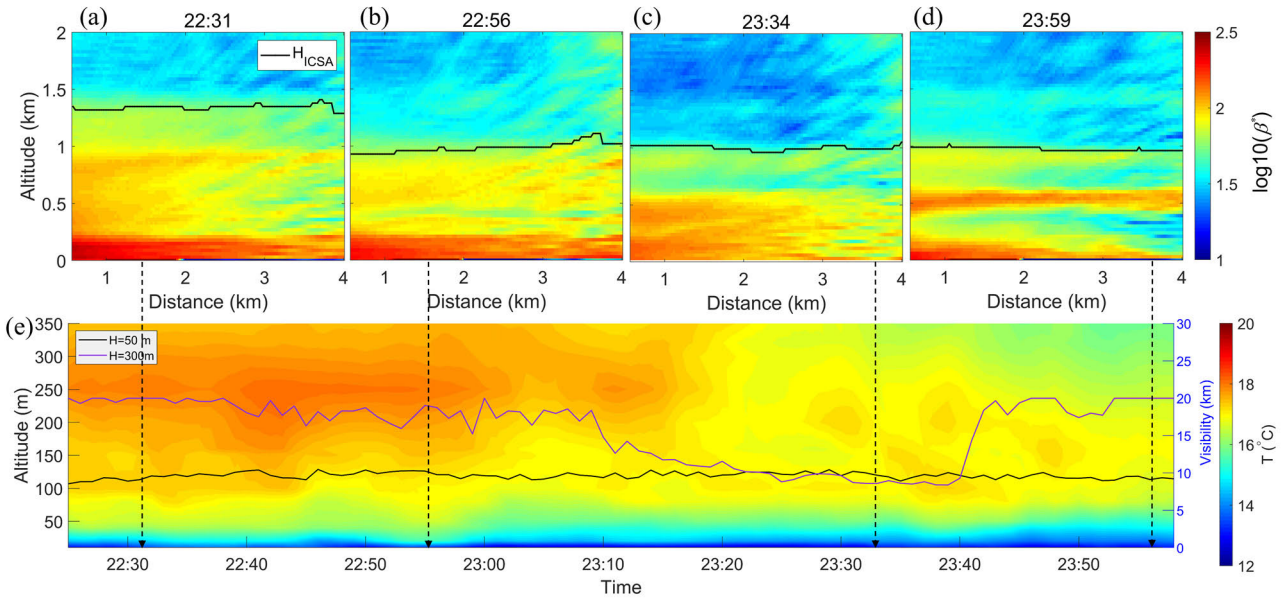


Fig. 14. (a)–(d) LiDAR observation images, meteorological tower observation temperature (color shading), and visibility at 50 m (black solid line) and 300 m (purple solid line) on the night of January 20, 2025. The black solid lines in (a)–(d) represent the ABLH identified by ICSA, and the black dashed lines indicate the times of the LiDAR images.

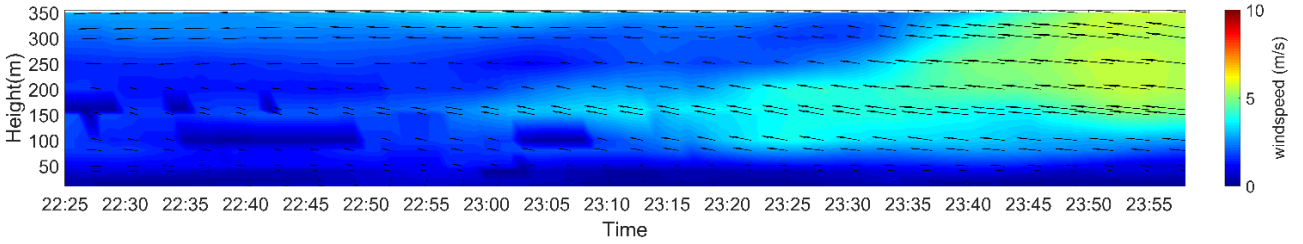


Fig. 15. Meteorological tower observation wind speed (color shading) and direction (arrows) on the night of January 20, 2025.

decreased to approximately 10 km, and recovered to 20 km after 23:43.

During this observation, the wind remained a stable easterly at speeds below 5 m/s, with no significant wind shear or laminar effects (see Fig. 15). Therefore, the anomalous LiDAR-observed aerosol dispersion under stable atmospheric conditions can be attributed to the evolution of the inversion layer rather than to wind dynamics. Under a stable nocturnal temperature inversion, aerosols are trapped within the layer due to suppressed vertical mixing, leading to elevated concentrations and persistent stratification. As the vertical temperature gradient changes, the inversion layer height decreases, disrupting the existing thermal equilibrium. The warmer air below gains buoyancy and rises, while the cooler air above descends to compensate, thereby triggering thermal convection. This process drives the vertical transport of aerosols, allowing them to cross the original boundary and disperse upward, thereby elevating the previously stable layer. After convective mixing, the atmosphere's thermal structure reestablishes equilibrium. Some aerosols remain suspended aloft, forming new layers that correspond to the temperature stratification. Near the surface, regions still under the inversion maintain their original high aerosol concentrations.

Due to the very low height of the near-surface temperature inversion, the proposed ICSA can capture the response of the shallow nocturnal atmospheric boundary layer to the evolution of the atmospheric thermal structure. This capability provides valuable support for detailed investigations of thermodynamically driven processes within the boundary layer.

V. Conclusion

The atmospheric boundary layer serves as a crucial interface for land–atmosphere interactions. High-resolution observation of the atmospheric boundary layer is not only fundamental to understanding its physical characteristics but also central to applied research in meteorology and air pollution. LiDAR, with its high temporal and spatial resolution, has become a powerful tool for atmospheric boundary layer detection. However, conventional vertically pointing LiDAR systems are inherently limited by a near-surface blind zone, which constrains the ability to capture the full vertical structure of the atmospheric boundary layer and poses challenges for detecting ABLH under different stability conditions.

To address these limitations, this study employs the RHI scanning mode of aerosol LiDAR for atmospheric boundary layer detection. In the RHI mode, LiDAR constructs the 2-D

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horizontal and vertical aerosol distribution, thereby revealing its internal stratification with improved clarity. Compared to the fixed vertical observation mode, this 2-D spatial perspective provides richer structural information and significantly enhances the robustness of ABLH determination. Leveraging this advantage, the ICSA is proposed to determine the ABLH from RHI 2-D imagery. The ICSA utilizes the Otsu adaptive thresholding technique for image segmentation. By defining appropriate classification intervals, it automatically partitions the atmosphere into distinct layers. Connected-component analysis is then employed to precisely extract the ground-contacting atmospheric boundary layer region and determine its height. The results of ABLH derived from the ICSA were compared and validated against those from the GM and STRATfinder methods. A prescribed altitude range was applied only to the GM and STRATfinder methods to exclude cloud interference, and the three methods show overall agreement. In addition, ICSA exhibits stronger spatial continuity. This confirms the effectiveness and robustness of the ICSA, overcoming the challenge of layer attribution during ABLH identification.

Based on the RHI full-height spatial detection technique, this study successfully captured the detailed evolution of the boundary layer structure under strong wind conditions and the development of a near-surface temperature inversion. The observations clearly reveal the wind-driven dynamics of the boundary layer. At night, laminar wind flow transported aerosols horizontally in the upper boundary layer. Instability waves formed but did not develop into turbulence, limiting vertical diffusion and keeping near-surface aerosols elevated. In addition, under stable atmospheric conditions, the evolution of the surface-based temperature inversion triggered intermittent turbulence, which redistributed the nocturnal aerosol structure. These results provide valuable in situ evidence for elucidating the coupled thermodynamic and dynamic response mechanisms within the boundary layer under complex meteorological conditions.

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